

Automated Classification of Acid and Non-Acid Gastro-oesophageal Reflux from Ambulatory pH Monitoring: A Comparison of Supervised Learning Algorithms

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Abstract—Gastro-oesophageal reflux disease (GERD) is common across Asian populations, and 24-hour ambulatory pH monitoring is still the diagnostic gold standard. Reading these long recordings by hand is slow and varies between clinicians, which is a good reason to look at automating the classification of reflux episodes. In this study I compare three supervised learning algorithms on oesophageal pH data. The three are k-Nearest Neighbours (k-NN), Support Vector Machines (SVM) and a feedforward Neural Network (NN), and I ran them on statistical features that I built by cutting the raw signal into short windows. Instead of judging the models on accuracy alone, I also looked at the precision and recall trade-off, the training and inference cost and how suitable each model would be for a resource-aware clinical setting such as Singapore. All three reached similar test accuracy, between 0.81 and 0.83, with ROC-AUC values from 0.87 to 0.91. k-NN gave the highest ROC-AUC (0.903) and was almost free to train and run. The neural network gave the best recall (0.788) and the best average precision (0.925). The RBF SVM gave the highest precision (0.931) but missed the most acid windows. The fact that three very different models land so close together, and that the simplest one holds its own, suggests that the variability features carry most of the signal and that light models are a sensible choice for real-time monitoring.

Index Terms—Supervised learning, gastro-oesophageal reflux, pH monitoring, neural networks, support vector machines, k-nearest neighbours, time-series classification.

I. INTRODUCTION

Gastro-oesophageal reflux disease (GERD) is one of the most common problems seen in gastroenterology, and its prevalence across Asia has been climbing for years. Population studies put symptom-based GERD in Eastern Asia at roughly 2.5 to 8.5 per cent, which is lower than the 10 to 20 per cent usually reported in Western countries, although the gap has been narrowing as diets and body weight change [1]–[3]. Asian patients also tend to show a different pattern, with proportionally more non-erosive reflux disease (NERD) than erosive disease, so it is worth checking how diagnostic tools behave specifically in this region [1], [4].

The reference standard for diagnosing GERD is ambulatory pH, or pH-impedance, monitoring. A sensor sits in the oesophagus and records pH for about 24 hours. The number clinicians care about most is the acid exposure time (AET), meaning the fraction of the recording spent below pH 4. The updated Lyon

Consensus treats an AET above 6 per cent as clear evidence of reflux disease and below 4 per cent as evidence against it [5], [6]. The trouble is that reading these traces takes time and different readers do not always agree, and recent work has shown that machine learning can help. Deep learning has been used to grade reflux oesophagitis from endoscopic images [7], and artificial intelligence is starting to automate parts of oesophageal motility and pH-impedance analysis [8].

Even so, directly classifying the pH signal itself with standard machine learning is still not well explored, and most of the work that does exist comes from Western cohorts or small single-centre datasets. That leaves a gap. There is little systematic comparison of the common supervised algorithms on pH recordings, and even less that keeps one eye on the practical limits of deploying such a tool in a developing part of Asia. To work towards filling this gap I train and compare three widely used classifiers, k-NN, SVM and a neural network, on windowed features taken from an acid and a non-acid pH recording. I deliberately look past headline accuracy and weigh up the precision and recall trade-off, the training and inference cost and, in the end, which model I would actually deploy for real-time monitoring in a setting like Singapore.

II. METHODOLOGY

A. Dataset Description

The data are two oesophageal pH recordings. One is labelled acid reflux (class 1) and the other non-acid reflux (class 0). Each is a physiological time-series sampled once per second. The timestamps go up in steps of 1000 ms, and with around 86,400 samples per class each recording covers a full 24-hour study. After parsing I was left with 86,197 valid acid readings and 86,400 non-acid readings, so the dataset is close to balanced. The acid file also held about 200 corrupt lines, which were dropped automatically at load time by forcing every field to a number and discarding anything that would not convert.

B. Exploratory Data Analysis

The two classes are almost perfectly balanced at the raw-sample level (Fig. 1).

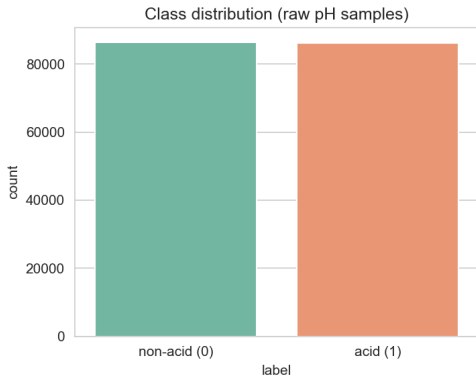


Fig. 1. Class distribution at the raw-sample level. The two classes are essentially balanced.

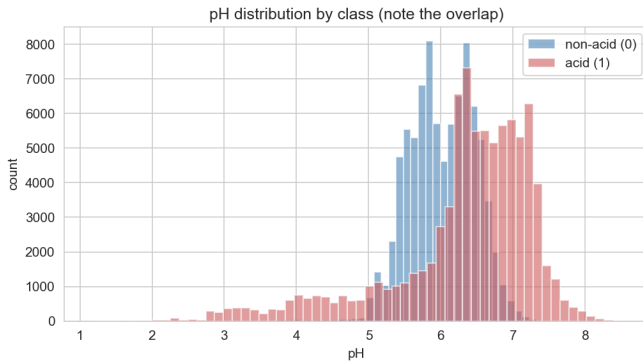


Fig. 2. pH distribution by class. The classes overlap heavily around pH 6, and the acid class shows a long acidic tail below pH 4.

The more important point is that the raw pH values overlap a great deal. Both classes sit around pH 6 (acid mean 6.32, non-acid mean 6.00), so a simple threshold on pH on its own cannot tell them apart (Fig. 2). What actually separates the classes is how much the signal moves. The acid recording is far more variable, with a standard deviation of 1.01 against 0.51, and it spends much more time in the acidic range. Its acid exposure time, the fraction of readings below pH 4, is 4.46 per cent, compared with only 0.47 per cent for the non-acid recording. In Lyon terms the acid class sits in the borderline to pathological AET band while the non-acid class is comfortably physiological.

A short raw slice of each class shows the difference plainly (Fig. 3). The acid trace swings between acidic dips and alkaline peaks, while the non-acid trace stays quiet around pH 6.

The anomaly check the brief asks for, pH below 0 or above 14, found nothing. I did notice a handful of physiologically odd readings below pH 2 (31 in the acid class and 120 in the non-acid class), which I put down to sensor artefacts. I kept them rather than deleting them, since they are rare and the windowing step below smooths over their effect.

One point on the clinical framing is worth making. True

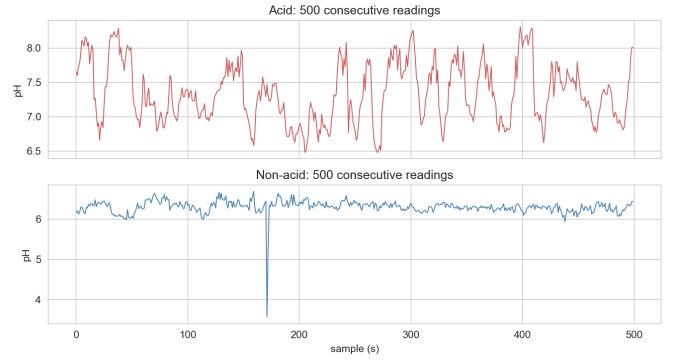


Fig. 3. Five hundred consecutive readings per class. The acid trace is volatile, while the non-acid trace stays around pH 6, apart from one sensor-artefact spike.

clinical acid reflux is defined at pH below 4, yet most of both recordings lies between pH 5 and 8. The labelling here is therefore better read as a *relative* acid versus non-acid split rather than the strict clinical cut-off, in line with the weakly-acidic (pH 4 to 7) and non-acid (above 7) categories used in impedance-pH work.

C. Feature Engineering

A single pH reading on its own says very little. The clinical signal lives in sustained drops or rises, so I cut each recording into non-overlapping windows and described every window with ten features. These are the mean, standard deviation, minimum, maximum, range and median of pH, the slope of a linear fit across the window and the proportion of readings above pH 7, below pH 6 and below pH 4. That last feature, `below_4_ratio`, is a per-window version of acid exposure time, and I added it because it maps so directly onto the clinical AET idea. A window never crosses from one recording to the other, so its label is never in doubt.

I built windowed datasets at three sizes, 30, 60 and 120 samples, which is 30, 60 and 120 seconds. These gave 5,753, 2,876 and 1,438 windows, each set close to balanced. Table I lists the class-conditional means of the 60-second features and backs up the picture from the EDA. The mean pH barely separates the classes and the slope is essentially zero for both, but the variability and tail features (`std_ph`, `range_ph`, `above_7_ratio` and `below_4_ratio`) differ sharply. In particular `above_7_ratio` jumps from 0.006 to 0.248 between the classes. This is hand-crafted time-series feature extraction, and libraries such as `tsfresh` generalise the same idea [9].

D. Preprocessing and Split

I took the 60-second dataset as my main modelling set. I chose 60 seconds because it gave a healthy number of windows while still covering a meaningful stretch of signal. I split it 80/20 with stratified sampling into 2,300 training and 576 test windows, then standardised the features with a `StandardScaler` that was fitted on the training data only

TABLE I
CLASS-CONDITIONAL FEATURE MEANS (60 s WINDOWS)

Feature	Non-acid (0)	Acid (1)
mean_ph	6.002	6.323
std_ph	0.128	0.244
range_ph	0.523	0.944
median_ph	6.012	6.326
slope	≈ 0	≈ 0
above_7_ratio	0.006	0.248
below_6_ratio	0.492	0.231
below_4_ratio	0.005	0.045

and then applied to both sets. Every model reused this one split so that the comparison stays fair. All random seeds, in Python, NumPy and TensorFlow, were fixed at 14 so the whole thing reproduces.

E. Machine Learning Algorithms

I built every model with scikit-learn and Keras rather than coding the algorithms from scratch. k-NN [10] was swept over $k \in \{1, 3, 5, 7, 11, 15, 21, 31, 51, 101\}$, over both the Euclidean ($p=2$) and Manhattan ($p=1$) metrics and over uniform and distance weighting. SVM [11] was tuned with `GridSearchCV` across two kernels, an RBF kernel (varying C and γ) and a polynomial kernel (varying C , γ and degree). The neural network is a feedforward Keras model. I compared a single-hidden-layer ReLU network, a two-layer ReLU network with dropout 0.3 and a three-layer Tanh network with $L2$ regularisation, then compared Adam against SGD with momentum on whichever of these did best. Dropout is a standard way to hold back over-fitting in networks like these [12].

F. Experimental Setup and Metrics

Model selection used 5-fold cross-validation on the training set only. The held-out test set was used once, at the very end. I report accuracy, precision, recall, F1-score and ROC-AUC, and I also record training and inference time as a rough measure of cost. For each winning model I produced a confusion matrix, and I plotted all of the models together on shared ROC and precision-recall axes. Everything runs under Python 3.12 with fixed seeds and reproduces on a standard Linux machine.

III. RESULTS

Table II gives the best test-set performance for each model. All four settings land in a narrow band of accuracy (0.809 to 0.826) and ROC-AUC (0.870 to 0.908), but they differ a lot in how they spread their errors and in how much they cost to run.

A. k-Nearest Neighbours

The best k-NN setting was $k=15$ with the Euclidean metric and uniform weighting, picked out by a cross-validated accuracy of 0.821. On the test set it reached accuracy 0.826, precision 0.873, recall 0.764, F1 0.815 and ROC-AUC 0.903, which is the highest AUC of any model. The accuracy against k curve (Fig. 4) is the classic bias and variance shape. At $k=1$

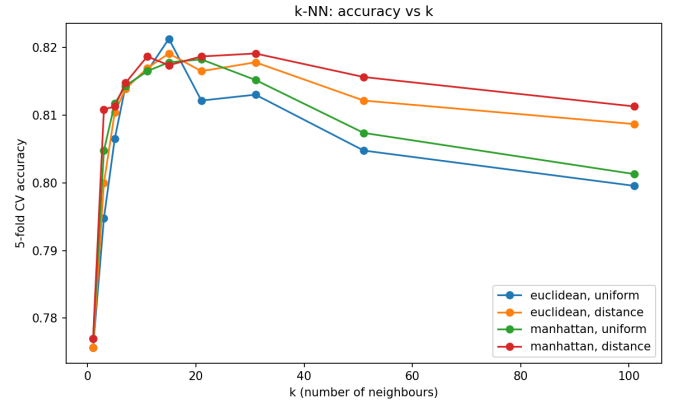


Fig. 4. k-NN cross-validated accuracy against k . Small k over-fits, performance peaks near $k=11$ to 15 , and large k over-smooths.

the model over-fits and the cross-validated accuracy is at its lowest. Performance then climbs to a peak around $k=11$ to 15 before drifting down again as very large k over-smooths the boundary and starts to underfit. The distance metric made little difference. Manhattan with distance weighting edged ahead at large k , but Euclidean with uniform weighting was best at the k I settled on.

The confusion matrix (Fig. 7, left) shows 68 missed acid windows against only 32 false positives, so the model is cautious. It under-calls acid rather than over-calling it. Both training and inference were almost instant, at about 34 ms and 33 ms.

B. Support Vector Machines

The RBF kernel beat the polynomial kernel. Its best setting ($C=100$, $\gamma=0.01$, cross-validated accuracy 0.814) gave test accuracy 0.823, precision 0.931, recall 0.698, F1 0.798 and ROC-AUC 0.887. The best polynomial model ($C=10$, degree 2, $\gamma=0.1$) trailed at accuracy 0.809 and AUC 0.870. The C and γ heatmap (Fig. 5) makes the tuning easy to read. At low C and low γ the model falls back to chance (0.501). Accuracy improves as C grows and γ takes middle values, peaking in the high- C , mid- γ corner, and the very largest γ values start to pull cross-validated accuracy down again as the boundary over-fits. Here C sets the trade-off between a wide margin and misclassification, and γ sets the kernel width, that is, how local the influence of each support vector is.

The RBF model gave the highest precision of any model (0.931, with only 15 false positives) but the lowest recall (0.698, with 87 missed acid windows), which makes it the most cautious classifier of the set (Fig. 7, centre). Training stayed cheap on this small dataset, at about 0.15 s, although SVM cost is known to climb steeply as the sample size grows.

C. Neural Network

Of the three architectures, the two-layer ReLU network with dropout gave the best cross-validated accuracy, ahead of the single-layer ReLU and the three-layer Tanh with $L2$. On that winning architecture Adam narrowly beat SGD with

TABLE II
BEST-MODEL PERFORMANCE ON THE HELD-OUT TEST SET (576 WINDOWS)

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Train (s)	Inference (s)
k-NN ($k=15$, Euclidean)	0.826	0.873	0.764	0.815	0.903	0.034	0.033
SVM (RBF, $C=100$, $\gamma=0.01$)	0.823	0.931	0.698	0.798	0.887	0.154	0.157
SVM (Poly, $C=10$, deg 2)	0.809	0.908	0.688	0.783	0.870	0.137	0.016
Neural Net (2-layer ReLU+drop, Adam)	0.819	0.841	0.788	0.814	0.908	10.56	0.139

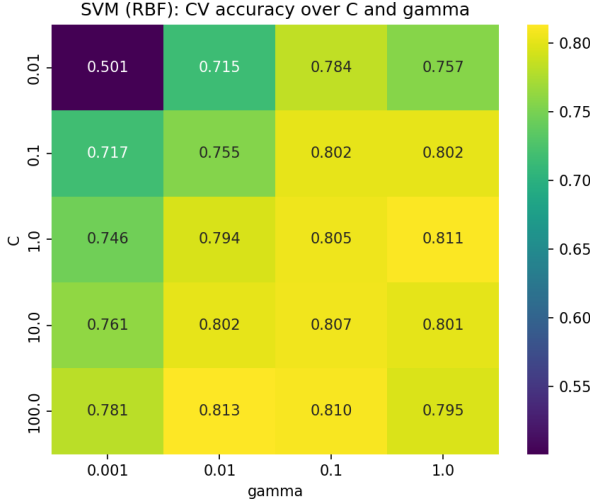


Fig. 5. SVM (RBF) cross-validated accuracy over C and γ . Low C and low γ collapses to chance, and the sweet spot lies at high C and moderate γ .

momentum. The final model reached test accuracy 0.819, precision 0.841, recall 0.788, F1 0.814 and ROC-AUC 0.908, the highest AUC and the highest average precision (0.925) of any model. The learning curves (Fig. 6) show training and validation accuracy moving together, with validation loss sitting at or below training loss thanks to the dropout and $L2$. There is no divergence, which points to good generalisation. The three-layer Tanh network showed a wider gap between training and validation, a mild sign of over-fitting that the regularised two-layer model managed to avoid.

The network also gave the best recall of all, missing only 61 acid windows (Fig. 7, right), at the cost of the slowest training, about 10.6 s, which is two to three orders of magnitude above the others.

D. Comparative Analysis

The confusion matrices for the three best models (Fig. 7) make the shared behaviour easy to see. Every model leans the same way, producing more false negatives, the missed acid windows in the lower-left, than false positives in the upper-right. Where they differ is the degree, and this ties back to the precision and recall figures in Table II.

Plotted on the same axes, the ROC curves (Fig. 8) sit close together, with k-NN and the neural network a little above the two SVM kernels.

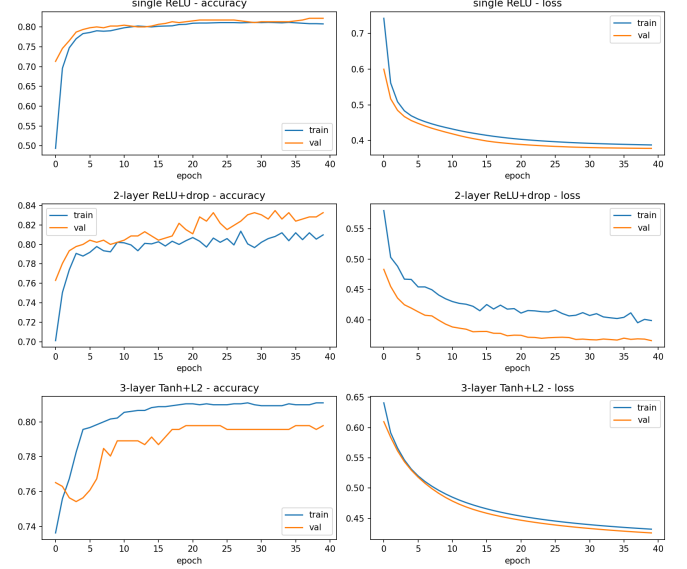


Fig. 6. Neural network learning curves. The regularised two-layer model tracks train and validation closely, while the three-layer Tanh network shows a wider gap.

TABLE III
EFFECT OF WINDOW SIZE (FIXED K-NN, $k=15$)

Window (s)	Accuracy	F1	ROC-AUC
30	0.798	0.789	0.886
60	0.826	0.815	0.903
120	0.826	0.820	0.912

The precision-recall curves (Fig. 9) tell much the same story, with the neural network taking the highest average precision (0.925).

Finally, the window-size experiment (Fig. 10, Table III), which held the best k-NN fixed and changed only the window length, shows that 30-second windows are clearly weaker (AUC 0.886), while 60 and 120 seconds level off on accuracy (0.826) with 120 seconds giving the best AUC (0.912). Longer windows help because AET-style reflux signals are by nature durational, but the gain flattens out.

IV. DISCUSSION

The result that stood out to me most is how closely the three algorithms agree. They land within about one percentage point of accuracy and within 0.04 of one another on ROC-AUC. To me this says the ceiling here is set by the data and not by the

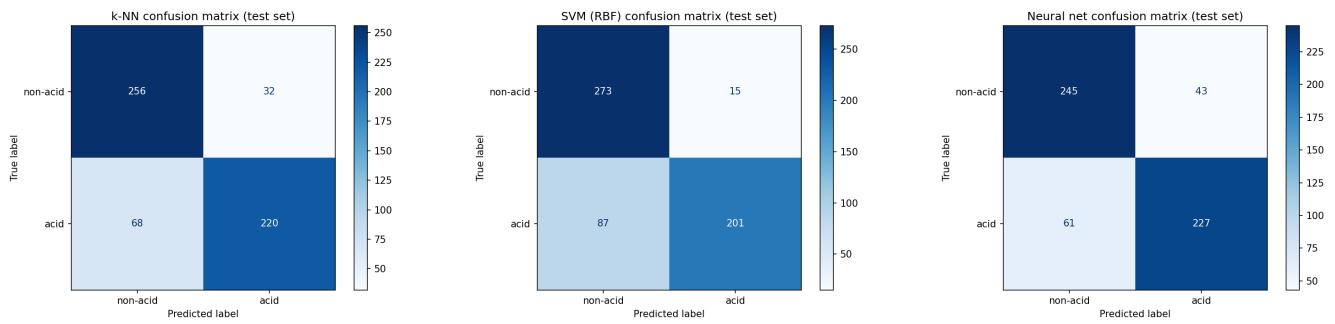


Fig. 7. Test-set confusion matrices for the best model of each type: k-NN (left), SVM with the RBF kernel (centre) and the neural network (right). All three make more false negatives (missed acid windows) than false positives, with the SVM the most cautious and the neural network the most sensitive.

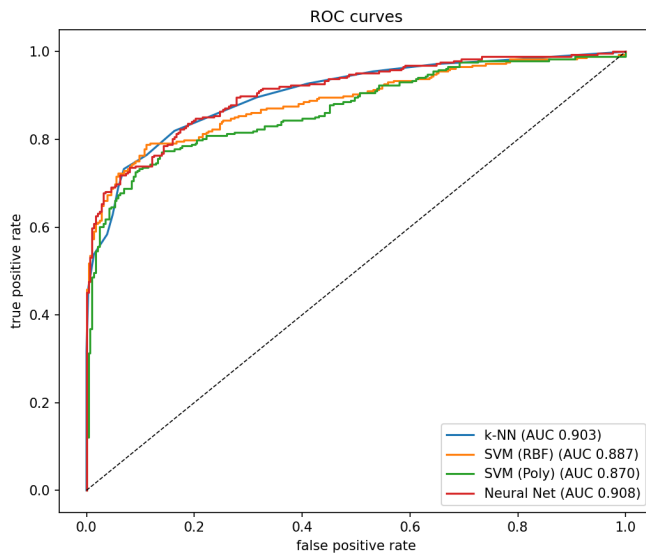


Fig. 8. ROC curves for all models. The curves sit close together, with k-NN and the neural network a little ahead.

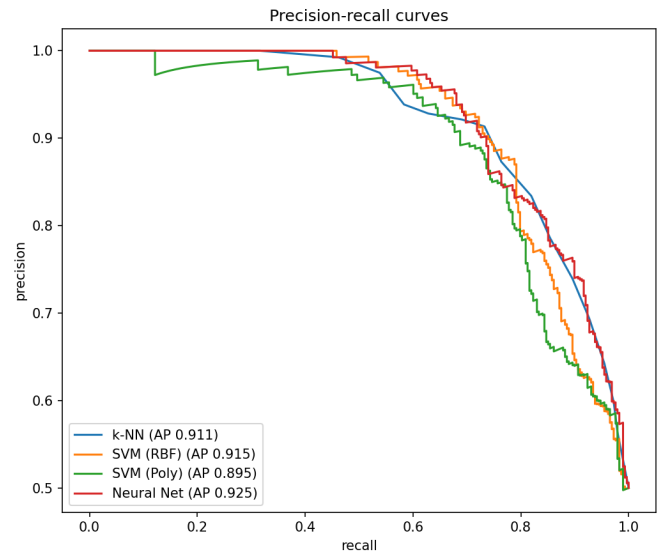


Fig. 9. Precision-recall curves. The neural network takes the highest average precision (0.925).

model. Because the raw pH distributions of the two classes overlap almost completely, no classifier can pull them apart from pH alone. What makes any classification possible is the engineered variability and tail features, and once those are in place a simple nearest-neighbour rule already captures most of what is there. A plateau like this is common in biomedical data, where measurement noise and natural variation between people cap how far accuracy can go.

The more useful way to separate the models is by how they get things wrong. Every model produced more false negatives than false positives, so each one under-calls acid reflux, but they do it to different degrees. The RBF SVM is the extreme. Its precision is excellent (0.931) but its recall is the worst (0.698). When it does flag a window as acid it is nearly always right, but it lets a lot of genuine acid windows slip past. The neural network sits at the other end, with the best recall and the fewest missed episodes. In a monitoring context this is not just a cosmetic difference. A missed acid episode and a false alarm carry different clinical costs, and a screening tool

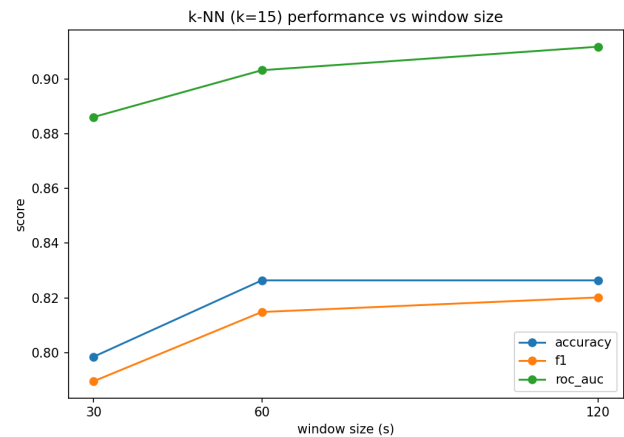


Fig. 10. Effect of window size on a fixed k-NN. Performance improves from 30 to 60 seconds and levels off by 120 seconds.

would usually want the higher recall of the network, while a confirmatory tool might prefer the precision of the SVM.

Each model also has its own profile. k-NN is the simplest and cheapest of the three. It gave the highest ROC-AUC, trained and predicted in milliseconds and is easy to interpret, though it is sensitive to k and over-fits badly when k is small. The SVM needed careful joint tuning of C and γ , and the heatmap shows how quickly it falls away from the sweet spot. It gave the best precision, but its training cost scales poorly with data size, which would matter if this were pushed out to many full 24-hour recordings. The neural network gave the best AUC, average precision and recall and generalised cleanly, but it took far longer to train and is the hardest to interpret. The window-size study adds a small practical lesson of its own. Sixty seconds is a good default, and 120 seconds squeezes out a little more discriminative power at the price of having fewer windows to train on.

For deployment in a resource-aware Asian setting such as Singapore, no single model wins on every front, so the choice comes down to what matters most. For real-time or on-device monitoring, or anywhere compute is limited, I would recommend k-NN as the default. It is essentially free to run, it is interpretable and it is competitive on every metric. Where catching as many reflux episodes as possible is the priority, and there is compute to spare, the neural network is the better pick for its higher recall and AUC. The SVM is the one to reach for when avoiding false alarms is the main concern. I will admit I went in expecting the neural network to win comfortably. In the end it led only on AUC, recall and average precision, while the much cheaper k-NN stayed right on its heels, which is a good reminder that the choice of model should follow the results and not an assumption about which one ought to be strongest.

A few limitations should temper all of this. The data are two individual recordings rather than a population sample, so the results speak to how separable the signal is and not to how well this would generalise across patients. The labelling is a relative acid versus non-acid split rather than the strict clinical pH below 4 threshold, and I chose to keep the sensor artefacts in. There is no external validation on an independent patient, and the feature set, while clinically motivated, is fairly small. Future work should test this on multi-patient population data, bring in pH-impedance channels and richer features and check whether it holds up across patients. Automated feature-extraction pipelines would also be worth trying, to lean less on hand-picked features.

V. CONCLUSION

I compared k-NN, SVM and a neural network for telling acid from non-acid gastro-oesophageal reflux using windowed pH features. All three reached similar test accuracy, between 0.81 and 0.83, with ROC-AUC values from 0.87 to 0.908. The neural network was strongest on AUC and average precision, k-NN followed closely at a tiny fraction of the cost and the RBF SVM was the most precise but the least sensitive. Variability and acid-exposure features, not raw pH,

carried the signal, and window lengths of 60 to 120 seconds worked best. The way such different models converge, and the strong showing of the simplest one, suggests that light and interpretable classifiers are a good fit for real-time reflux monitoring in resource-aware clinical settings, with the higher-recall neural network held in reserve for cases where missing an episode is the greater risk.

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